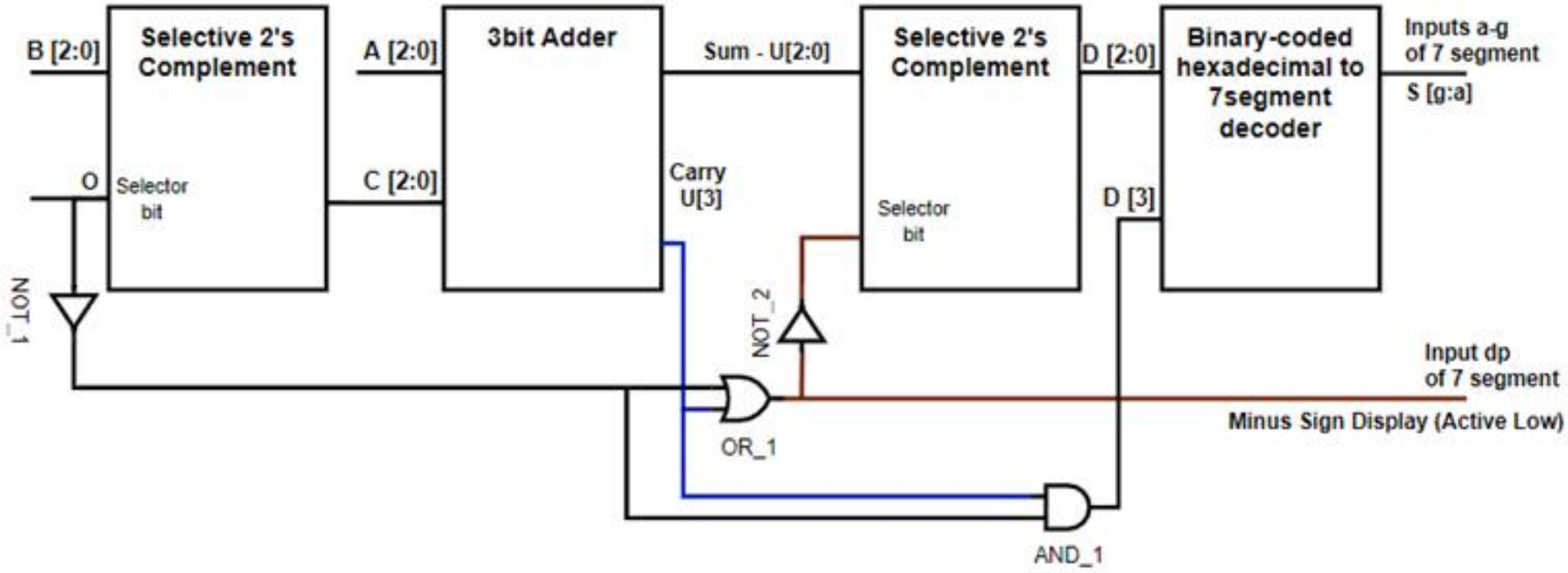
# DLD Lab 06
# Seven Segment Decoder Design

# Add/Subtract Module

## Lab 06, 07, 08

1 = Subtraction || Find 2's complement
0 = Addition || Pass through

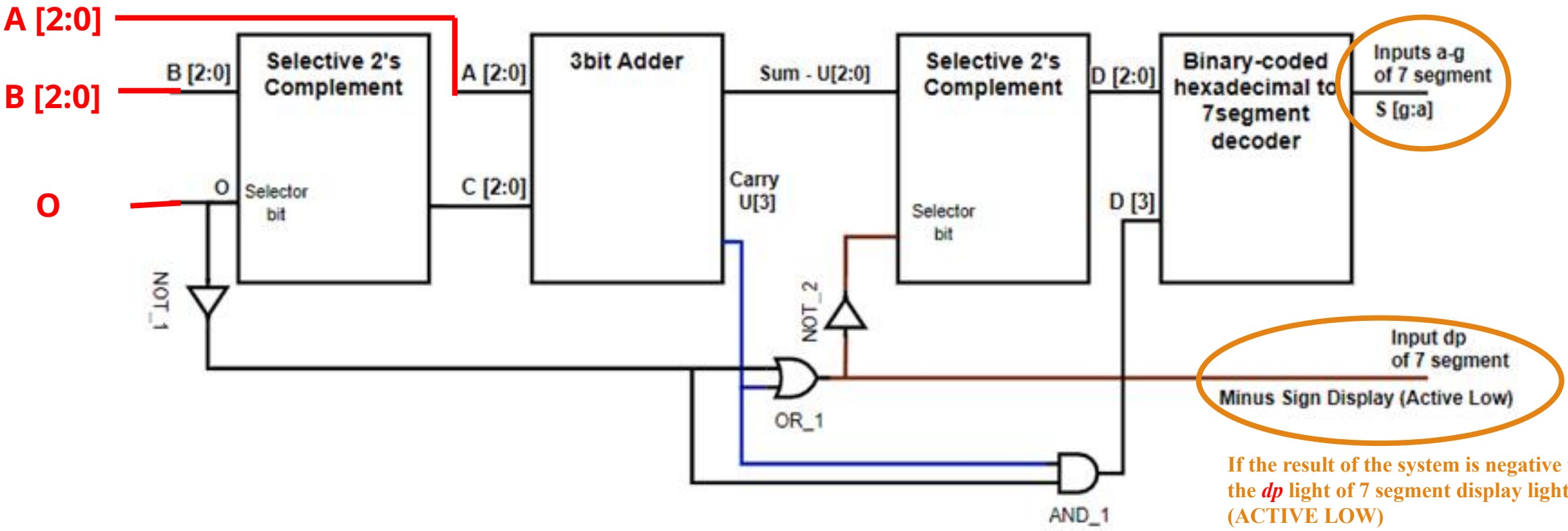
- System adds/subtracts two 3-bit binary numbers **A & B.**
- **O** is another 1 bit input that determines if the system will add/subtract**.**



# Add/Subtract Module

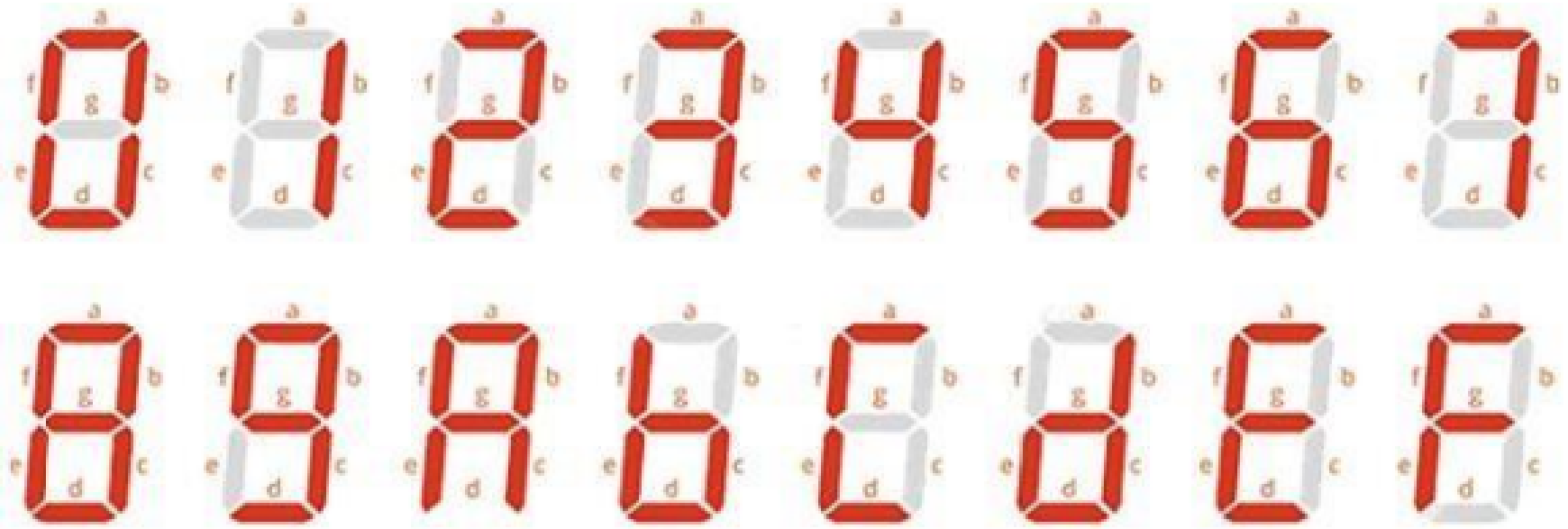
1 = Subtraction || Find 2's complement
0 = Addition || Pass through

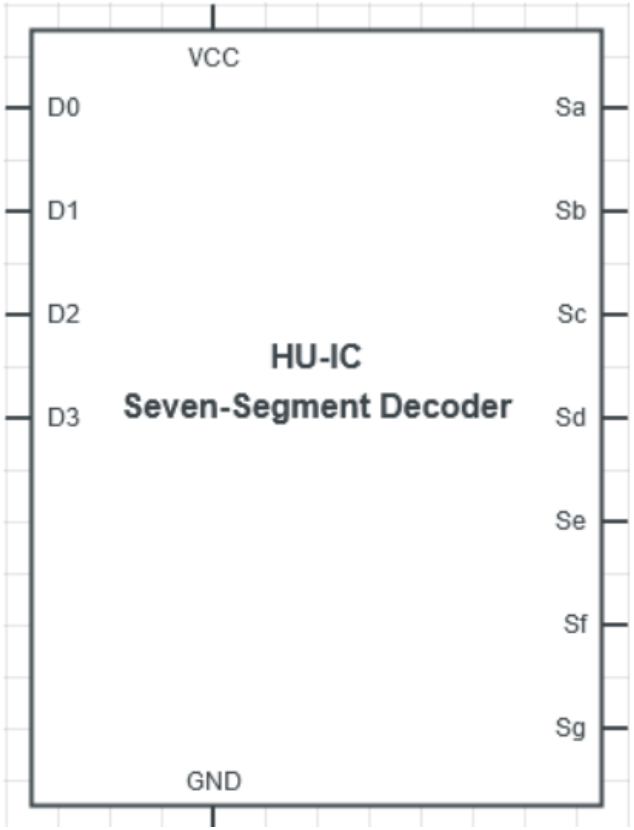
- System generates the appropriate signals for 7 segment display to show the result of addition/subtraction as a hexadecimal digit.
- Common Anode (Active Low)




If the result of the system is negative then the ***dp*** light of 7 segment display lights up (ACTIVE LOW)

# Seven Segment Display






- 7 segment decoder module replaces 7447 IC that takes 4 inputs (BCD) and decodes it into seven outputs.
- Decoder takes 4 input bits (D3-D0) representing binary coded hexadecimal number between 0 & F, and produces seven output bit s(Sg-Sa) – lights up appropriate segments of 7 segment display.

# Seven Segment Display

## Common Anode – Active Low

0 0 0 1

$D_3$ $D_2$ $D_1$ $D_0$


1 1 1 1 0 0 1

g f e d c b a


0 0 0 1 0 0 0

g f e d c b a

1 0 1 0

$D_3$ $D_2$ $D_1$ $D_0$

## Example 1: Add 2 and 3 in decimal and display its result on seven segment display.

## Operation: 2 + 3 = 5

A = 3'b010

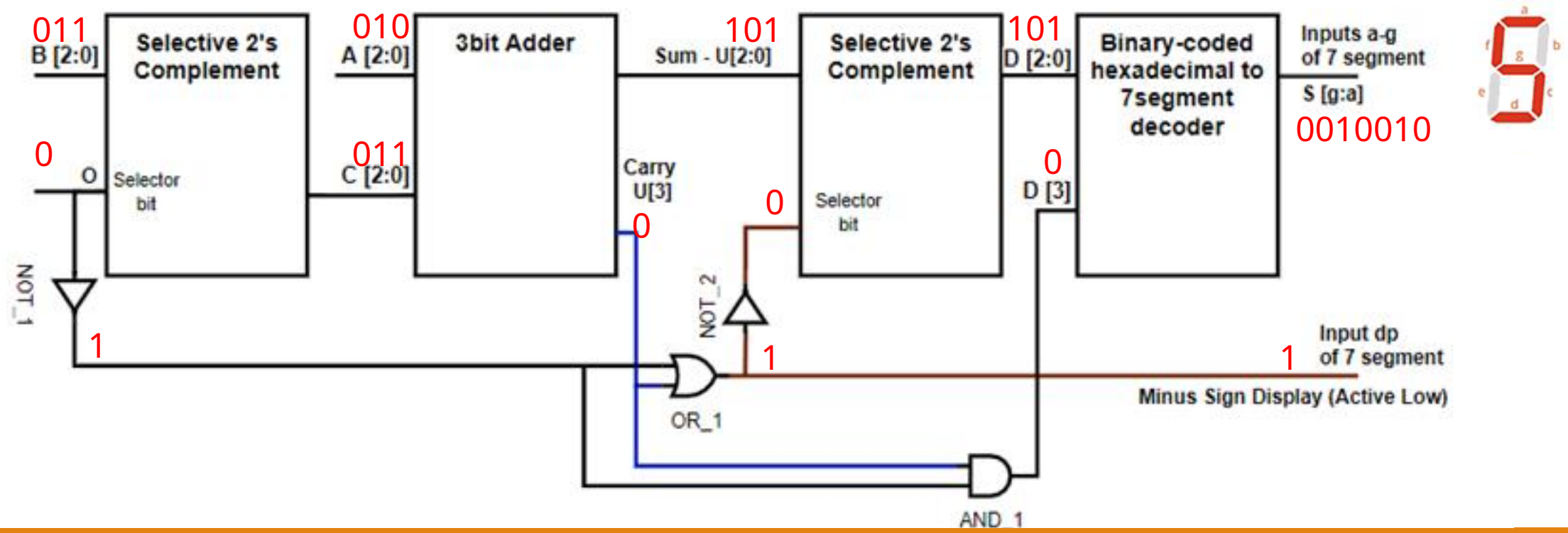
B = 3'b011

O = 0

1 = Subtraction || Find 2's complement

0 = Addition || Pass through

| Hex Digit | Binary-coded Hexadecimal/ Inputs to decoder | | | | Outputs of the decoder | | | | | | |
|---|---|---|---|---|---|---|---|---|---|---|---|
| | $D_3$ | $D_2$ | $D_1$ | $D_0$ | $S_g$ | $S_f$ | $S_e$ | $S_d$ | $S_c$ | $S_b$ | $S_a$ |
| 0 | 0 | 0 | 0 | 0 | 1 | 0 | 0 | 0 | 0 | 0 | 0 |
| 1 | 0 | 0 | 0 | 1 | 1 | 1 | 1 | 1 | 0 | 0 | 1 |
| 2 | 0 | 0 | 1 | 0 | 0 | 1 | 0 | 0 | 1 | 0 | 0 |
| 3 | 0 | 0 | 1 | 1 | 0 | 1 | 1 | 0 | 0 | 0 | 0 |
| 4 | 0 | 1 | 0 | 0 | 0 | 0 | 1 | 1 | 0 | 0 | 1 |
| 5 | 0 | 1 | 0 | 1 | 0 | 0 | 1 | 0 | 0 | 1 | 0 |
| 6 | 0 | 1 | 1 | 0 | 0 | 0 | 0 | 0 | 0 | 1 | 0 |



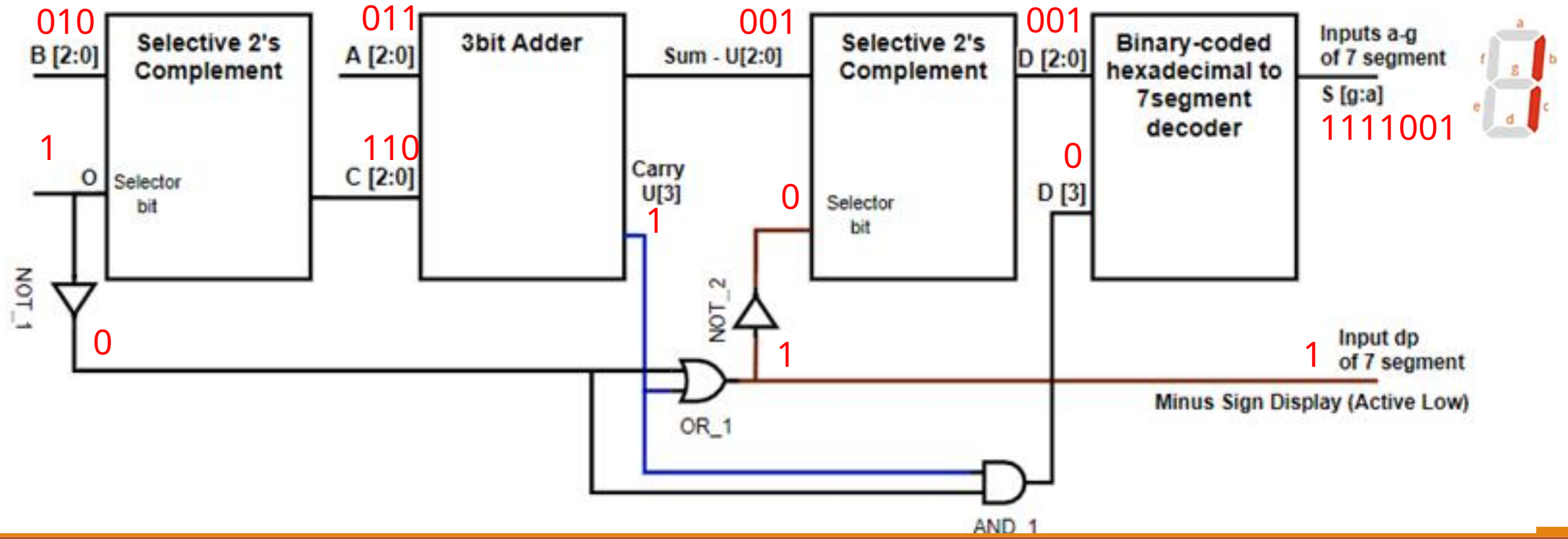
## Example 2: Subtract 2 from 3 in decimal and display its result on seven segment display.

## Operation: 3 – 2 = 1

A = 3'b011
B = 3'b010
O = 1

1 = Subtraction || Find 2's complement
0 = Addition || Pass through

| Hex Digit | Binary-coded Hexadecimal/ Inputs to decoder | | | | Outputs of the decoder | | | | | | |
|---|---|---|---|---|---|---|---|---|---|---|---|
| | $D_3$ | $D_2$ | $D_1$ | $D_0$ | $S_g$ | $S_f$ | $S_e$ | $S_d$ | $S_c$ | $S_b$ | $S_a$ |
| 0 | 0 | 0 | 0 | 0 | 1 | 0 | 0 | 0 | 0 | 0 | 0 |
| 1 | 0 | 0 | 0 | 1 | 1 | 1 | 1 | 1 | 0 | 0 | 1 |
| 2 | 0 | 0 | 1 | 0 | 0 | 1 | 0 | 0 | 1 | 0 | 0 |
| 3 | 0 | 0 | 1 | 1 | 0 | 1 | 1 | 0 | 0 | 0 | 0 |
| 4 | 0 | 1 | 0 | 0 | 0 | 0 | 1 | 1 | 0 | 0 | 1 |
| 5 | 0 | 1 | 0 | 1 | 0 | 0 | 1 | 0 | 0 | 1 | 0 |
| 6 | 0 | 1 | 1 | 0 | 0 | 0 | 0 | 0 | 0 | 1 | 0 |



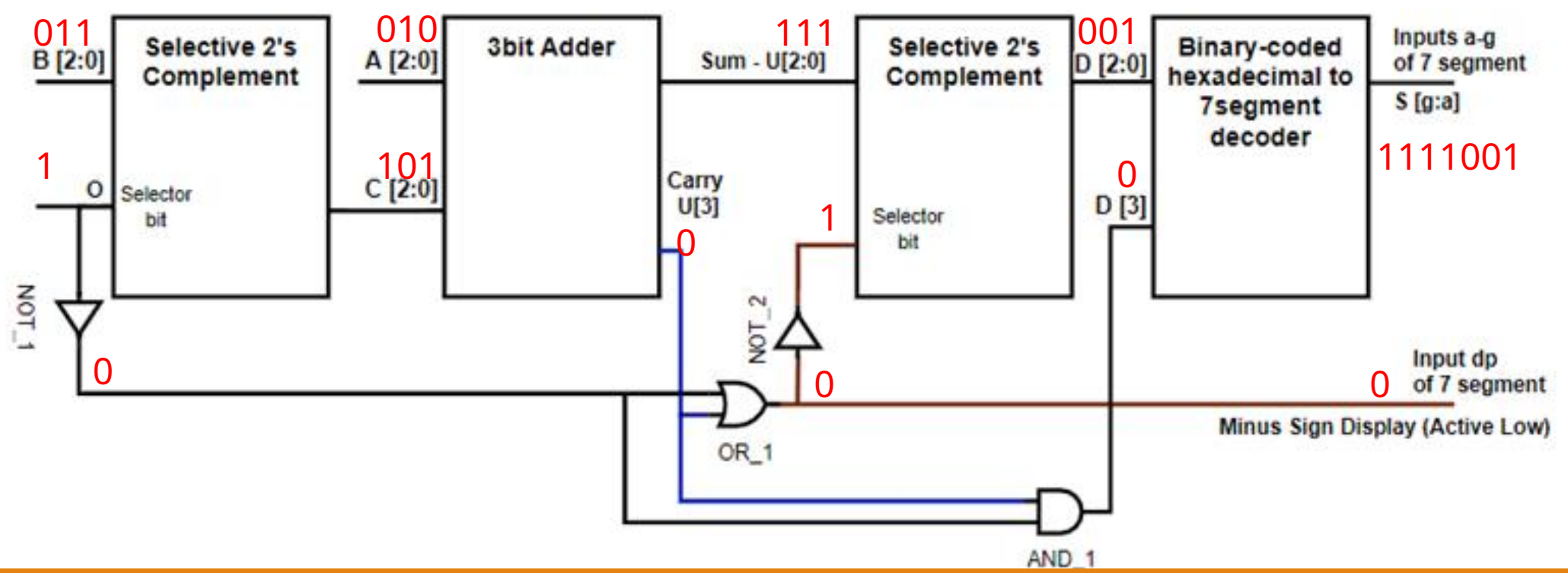
## Example 3: Subtract 3 from 2 in decimal and display its result on seven segment display.

## Operation: 2 – 3 = -1

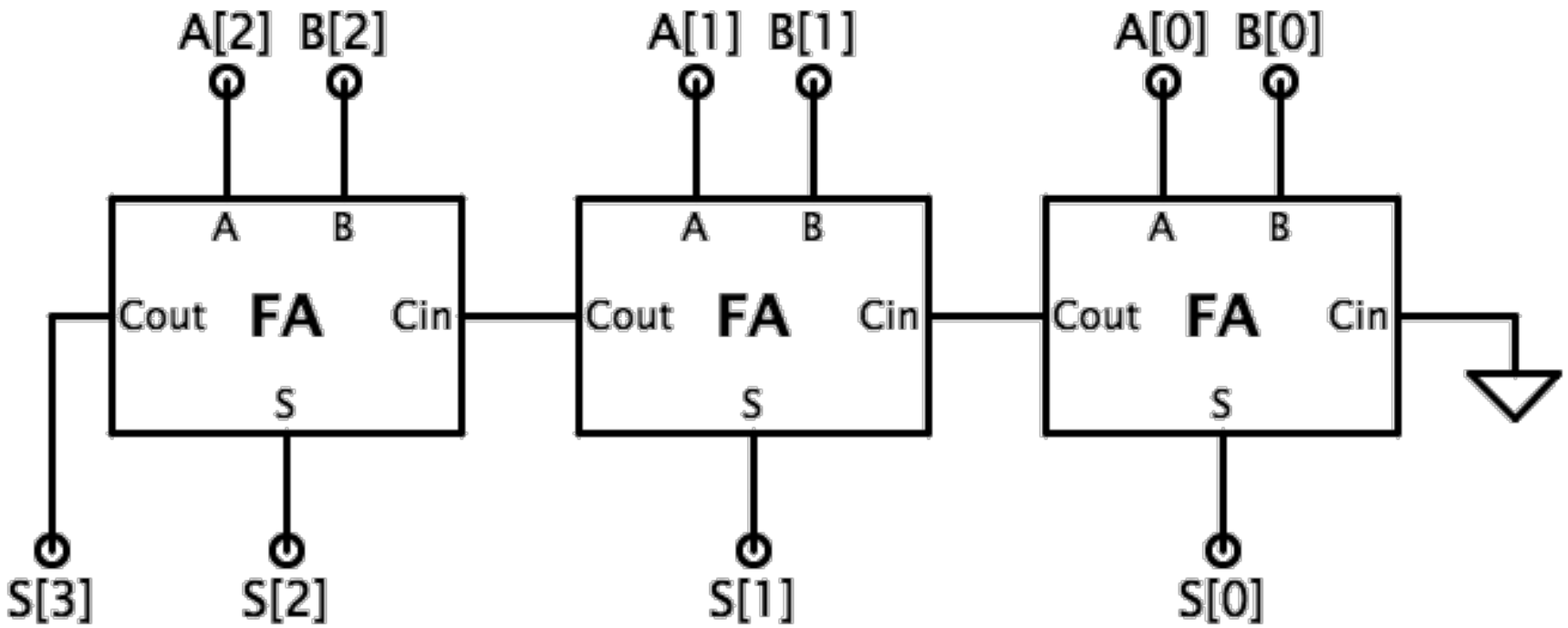
A = 3'b010
B = 3'b011
O = 1

1 = Subtraction || Find 2's complement
0 = Addition || Pass through

| Hex Digit | Binary-coded Hexadecimal/ Inputs to decoder | | | | Outputs of the decoder | | | | | | |
|---|---|---|---|---|---|---|---|---|---|---|---|
| | $D_3$ | $D_2$ | $D_1$ | $D_0$ | $S_g$ | $S_f$ | $S_e$ | $S_d$ | $S_c$ | $S_b$ | $S_a$ |
| 0 | 0 | 0 | 0 | 0 | 1 | 0 | 0 | 0 | 0 | 0 | 0 |
| 1 | 0 | 0 | 0 | 1 | 1 | 1 | 1 | 1 | 0 | 0 | 1 |
| 2 | 0 | 0 | 1 | 0 | 0 | 1 | 0 | 0 | 1 | 0 | 0 |
| 3 | 0 | 0 | 1 | 1 | 0 | 1 | 1 | 0 | 0 | 0 | 0 |
| 4 | 0 | 1 | 0 | 0 | 0 | 0 | 1 | 1 | 0 | 0 | 1 |
| 5 | 0 | 1 | 0 | 1 | 0 | 0 | 1 | 0 | 0 | 1 | 0 |
| 6 | 0 | 1 | 1 | 0 | 0 | 0 | 0 | 0 | 0 | 1 | 0 |

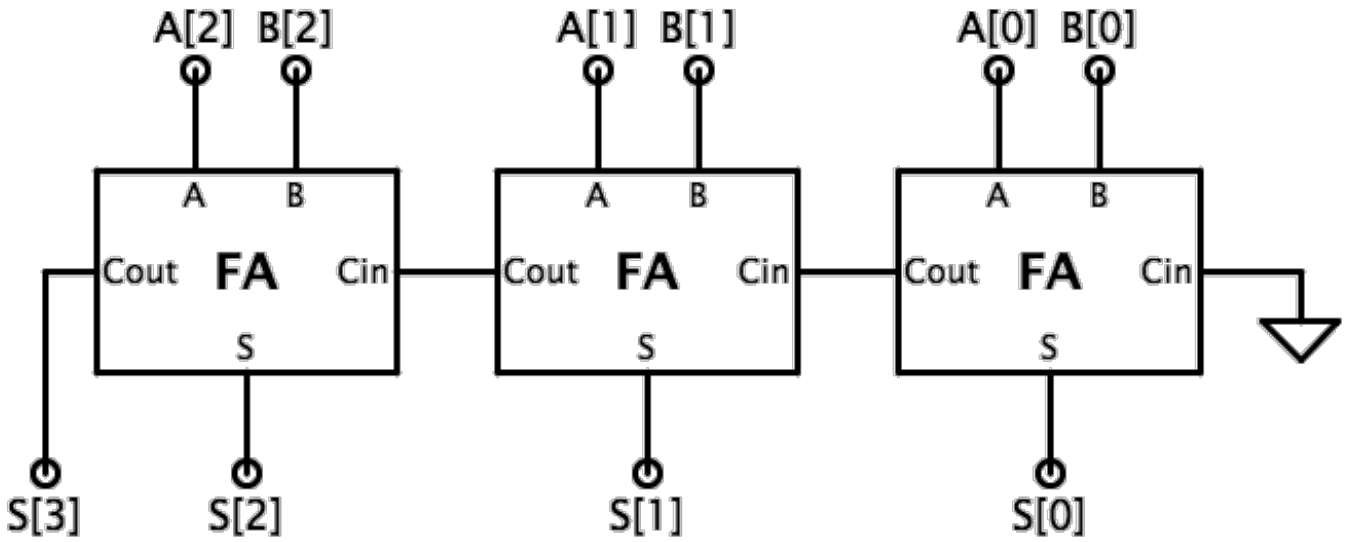


# Task c: 3 bits Full Adder



# Task c: 3 bits Full Adder

- Use the code provided in manual.
- Create 2 separate design files, FA using HA and 3-bit full adder using previous modules.



# Task Guidelines

***Task A.*** The similar **test cases** performed earlier is your task a (try a subtraction case with a negative output) (15 minutes)

***Task B.*** **Write code for 7 segment decoder**

- Fill out the truth table (15 minutes)
- Use online k-map generator to find Boolean expression (20 minutes)
- Module & Testbench (40 minutes)

***Task C.*** **Create a 3 bits Full Adder module**

- Use provided code to write full adder code – test if you wish. Testbench is not required
- Cascade 3 Full Adders to design a 3-bit Full Adder and test your module (30 minutes) – code is given
- Test cases: 2+1, 3+7, 3+3

***Both of these codes from Task b & c will be used in Lab 07 & 08!***

# Common Errors in Vivado

**Simulation – error**

- Close previous simulation and re-run
- Same step for the schematic

**Testing multiple testbenches**

- Right click the test module (you intend to simulate) click on Set as top and then Run Simulation

**Always check ‘Message’ tab for any error in the code**

# K-MAP Simplifier

- Choose the number of variable based on the number of inputs.
- For **each segment output (Sa through Sg)** choose the signal (0 or 1) for all possible combinations of inputs >> *A,B,C,D = D3, D2, D2, D0.*
- Create Boolean Expression for Sa through Sg outputs
- **Notation in Verilog:**
  **D3' D2' D1' D0 = ( ~D[3] & ~D[2] & ~D[1] & D[0] )**

Or = | **not** +

Not = ~

And = &

| Hex Digit | Binary-coded Hexadecimal/ Inputs to decoder | | | | Outputs of the decoder | | | | | | |
|---|---|---|---|---|---|---|---|---|---|---|---|
| | $D_3$ | $D_2$ | $D_1$ | $D_0$ | $S_g$ | $S_f$ | $S_e$ | $S_d$ | $S_c$ | $S_b$ | $S_a$ |
| 0 | 0 | 0 | 0 | 0 | 1 | 0 | 0 | 0 | 0 | 0 | 0 |
| 1 | 0 | 0 | 0 | 1 | 1 | 1 | 1 | 1 | 0 | 0 | 1 |
| 2 | 0 | 0 | 1 | 0 | 0 | 1 | 0 | 0 | 1 | 0 | 0 |
| 3 | 0 | 0 | 1 | 1 | 0 | 1 | 1 | 0 | 0 | 0 | 0 |
| 4 | 0 | 1 | 0 | 0 | 0 | 0 | 1 | 1 | 0 | 0 | 1 |
| 5 | 0 | 1 | 0 | 1 | 0 | 0 | 1 | 0 | 0 | 1 | 0 |
| 6 | 0 | 1 | 1 | 0 | 0 | 0 | 0 | 0 | 0 | 1 | 0 |

Link:
K-map simplifier

**D3 D2 D1 D0**
**A B C D**

# Announcements

- Team Formation – Number of groups in the section T2 = 7 at max (4-5 students each)
- In case anyone has not joined the group by tonight – will be randomly enrolled in a group (which has less than 4 members)
- Project Proposal Template & Sample will be released in the following week ON LMS